Rain- Based Traffic Alert System

Team No: 6

Team Name: Nova

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MileStone 2: Dataset Generation and ML model Development

1. Introduction

The second milestone focuses on building the data-driven component of the Intelligent Rain-Based Traffic Alert System. This includes generating a dataset from simulated sensor values and applying machine learning techniques to classify road safety conditions. The goal is to improve prediction accuracy and enable intelligent decision-making.

2. Dataset Generation

The dataset is synthetically generated using simulated sensor values obtained from the IoT system. Since real-time large-scale data collection is not feasible at this stage, random data generation is used to create multiple scenarios.

Features Used:

- Rain (0-4095) - Rain intensity
- Traffic (0 or 1) - Vehicle presence
- Distance (cm) - Distance between vehicles
- Temperature (°C) - Environmental condition
- Humidity(%) - Moisture level in air
- Visibility (low/medium/high or numeric)- Road visibility
- Speed(km/h) - Vehicle speed
- Risk - Output label (SAFE, CAUTION, HIGH, CRITICAL)
- Location (Areas in Bengaluru)
- Patrol Unit (A, B, C, D, E)
- Phone Number

- Random values are generated for each feature
- Different combinations simulate real-world conditions

2.1 Dataset Creation Method

- Each data point is labeled based on predefined rules
- Dataset is stored in CSV format for ML processing

2.2 Head of Dataset

```
 Dataset successfully created!
Total rows: 1500

Sample data:
   Rain  Traffic  Distance  Temperature  Humidity  Visibility  Speed  \
0  2034        0       142           35        99          80     72
1  4075        0        88           25       100          42     68
2  4094        0       154           36        44          76     89
3  1881        1       166           31        57          75     38
4   557        1        60           29        73          87     25

     Location    Patrol_Unit Phone_Number     Risk    Rain_mm   Rain_cm
0  BTM Layout  Patrol Unit A   9918738587     High  24.835165  2.483516
1   Jayanagar  Patrol Unit B   9138789585     High  49.755800  4.975580
2   Yelahanka  Patrol Unit A   9698345375     High  49.987790  4.998779
3  Whitefield  Patrol Unit D   9868353176  Caution  22.967033  2.296703
4  BTM Layout  Patrol Unit E   9680338194     Safe   6.800977  0.680098
```

3. Risk Classification Logic

1. CRITICAL : It combines High rain , traffic present and very low distance
2. HIGH : Medium rain or moderate distance
3. CAUTION : Low rain and safe distance
4. SAFE : No traffic and normal conditions

4. Machine Learning Model

Models Used:

1. Decision Tree

- Rule-based classification
- Easy to interpret

2. Logistic Regression

- Statistical model for classification

- Fast and efficient
- Works well with structured data

3. Support Vector Classifier (SVC)

- Separates data using optimal boundaries
- Performs well for complex classification problems

4. Radom Forest

- Ensemble of multiple trees
- High accuracy and robustness

5. Hybrid Model (Voting Classifier)

- Combines logistic regression, SVC and Decision Tree Classifier.

4.1 Model Workflow

1. Load dataset
2. Split into training and testing data
3. Train ML model
4. Predict risk levels
5. Evaluate accuracy

5. Alert System

Once the model has classified the risk as Critical, High, Caution and Safe, then we have taken the count of rows which falls under Critical and High, then we will see the locations which have high and critical risk and then since we have the feature called phone numbers of that respective location we will send alert messages to that particular patrol so that they can take the actions based on that like telling the passengers to change to another route which make cause less number of accidents

5.1 Code for Alert System

```
df = pd.read_excel("rain_traffic_dataset.xlsx")

print("Dataset Loaded!\n")

print(" CLASS DISTRIBUTION:\n")
class_counts = df['Risk'].value_counts()

print(class_counts)
```

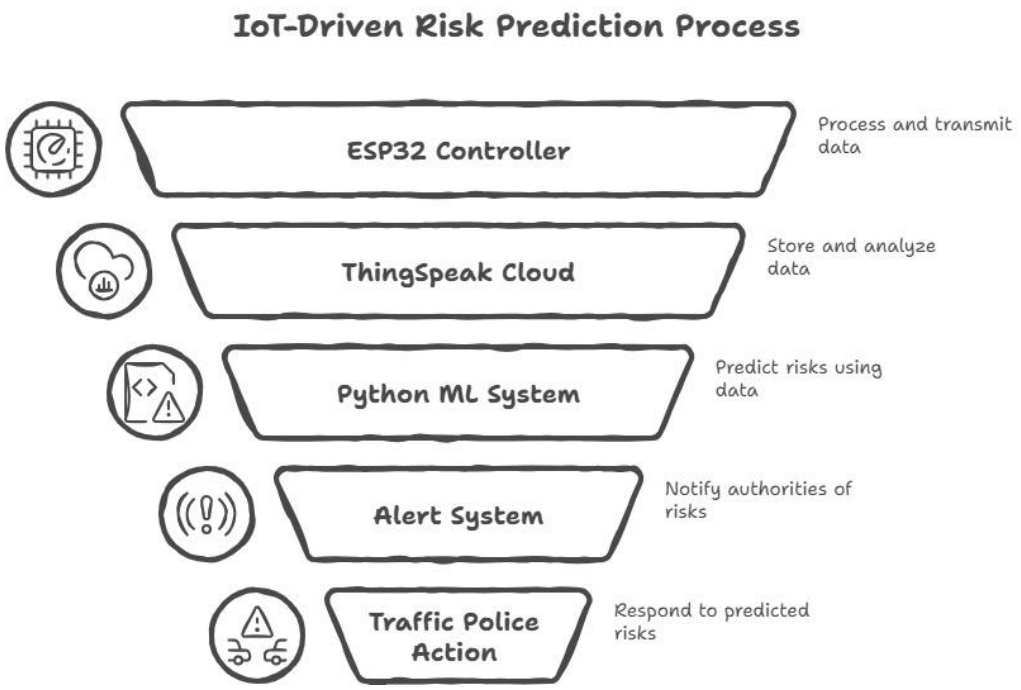
```
Dataset Loaded!

 CLASS DISTRIBUTION:

Risk
High        991
Caution     313
Safe        176
Critical     20
Name: count, dtype: int64
```

Fig 5.1 Count of Classifications

6. Data Storage in Excel

The generated dataset is stored in an Excel format to enable easy visualization, analysis, and sharing. Excel is used because it provides a simple tabular structure and supports integration with machine learning tools.

7. Model Workflow



8. Code

8.1DataSet Creation and Storing in Excel Automatically

```
import pandas as pd

import random

num_rows = 1500

data = []

locations = [

    "Whitefield",

    "Electronic City",

    "MG Road",

    "BTM Layout",

    "Yelahanka",

    "Indiranagar",

    "Marathahalli",

    "Hebbal",

    "KR Puram",

    "Jayanagar"

]

patrol_units = [

    "Patrol Unit A",

    "Patrol Unit B",

    "Patrol Unit C",

    "Patrol Unit D",

    "Patrol Unit E"

]

for i in range(num_rows):

    rain = random.randint(0, 4095)

    traffic = random.randint(0, 1)

    distance = random.randint(5, 200)
```

```
    temperature = random.randint(20, 40)
    humidity = random.randint(40, 100)
    visibility = random.randint(10, 100)
    speed = random.randint(20, 100)
    location = random.choice(locations)
    patrol = random.choice(patrol_units)
    # Generate random Indian phone number
    phone = "9" + "".join([str(random.randint(0,9)) for _ in range(9)])
    if rain > 3000 and distance < 20 and traffic == 1:
        risk = "Critical"
    elif rain > 2000 or distance < 40 or visibility < 30:
        risk = "High"
    elif rain > 1000 or speed > 80:
        risk = "Caution"
    else:
        risk = "Safe"
    data.append([
        rain, traffic, distance, temperature,
        humidity, visibility, speed,
        location, patrol, phone, risk
    ])
df = pd.DataFrame(data, columns=[
    "Rain",
    "Traffic",
    "Distance",
    "Temperature",
    "Humidity",
    "Visibility",
    "Speed",
    "Location",
```

```
    "Patrol_Unit",
    "Phone_Number",
    "Risk"
])
df['Rain_mm'] = (df['Rain'] / 4095) * 50
df['Rain_cm'] = df['Rain_mm'] / 10
df.to_excel("rain_traffic_dataset.xlsx", index=False)
df.to_csv("rain_traffic_dataset.csv", index=False)
print(" Dataset successfully created!")
print("Total rows:", len(df))
print("\nSample data:")
print(df.head())
```

8.2 Random Forest Model Training Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.ensemble import RandomForestClassifier
df = pd.read_excel("rain_traffic_dataset.xlsx")
print("Dataset Loaded!\n")

le = LabelEncoder()
df['Risk'] = le.fit_transform(df['Risk'])
print("\nEncoded Classes:", list(le.classes_))

X = df[['Rain', 'Traffic', 'Distance', 'Temperature',
        'Humidity', 'Visibility', 'Speed']]
y = df['Risk']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
```

```
)model = RandomForestClassifier(
    n_estimators=100,     # number of trees
    max_depth=10,         # control complexity
    random_state=42
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
acc = accuracy_score(y_test, y_pred)

print("\n🚀 RANDOM FOREST MODEL")
print("Accuracy: {:.2f}%".format(acc * 100))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

8.3 Hybrid Model Code(Decision Tree, Logistic Regression and SVC)

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.ensemble import VotingClassifier
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
df = pd.read_excel("rain_traffic_dataset.xlsx")
print("Dataset Loaded!\n")
le = LabelEncoder()
df['Risk'] = le.fit_transform(df['Risk'])
print("\nEncoded Classes:", list(le.classes_))
X = df[['Rain', 'Traffic', 'Distance', 'Temperature',
        'Humidity', 'Visibility', 'Speed']]
```

```
y = df['Risk']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model1 = LogisticRegression(max_iter=1000)
model2 = DecisionTreeClassifier(max_depth=5)
model3 = SVC(probability=True)
hybrid_model = VotingClassifier(
    estimators=[
        ('lr', model1),
        ('dt', model2),
        ('svm', model3)
    ],
    voting='soft'
)

hybrid_model.fit(X_train, y_train)
y_pred = hybrid_model.predict(X_test)
acc = accuracy_score(y_test, y_pred)
print("\n HYBRID MODEL")
print("Accuracy: {:.2f}%".format(acc * 100))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

9. Results and Outcomes

```
Dataset Loaded!


Encoded Classes: ['Caution', 'Critical', 'High', 'Safe']

 RANDOM FOREST MODEL
Accuracy: 99.67%

Confusion Matrix:
[[ 63   0   0   0]
 [  0   2   1   0]
 [  0   0 193   0]
 [  0   0   0  41]]
```

Fig 9.1 Random Forest Accuracy and Confusion Matrix

```
 HYBRID MODEL
Accuracy: 98.67%

Confusion Matrix:
[[ 60   0   1   2]
 [  0   3   0   0]
 [  0   0 192   1]
 [  0   0   0  41]]
```

Fig 9.1 Hybrid Model Accuracy and Confusion Matrix

10. Comparative Study

The comparative study of different machine learning models highlights variations in their performance for traffic risk prediction. Support Vector Machine (SVM), Logistic Regression, and Decision tree classifier combined together as a Hybrid model has achieved accuracy of (98.67%) indicating improved performance in handling the dataset. Random Forest has achieved an accuracy of (99.67%) which shows that the data is overfitting. So, we conclude that the Hybrid model has provided the highest accuracy.

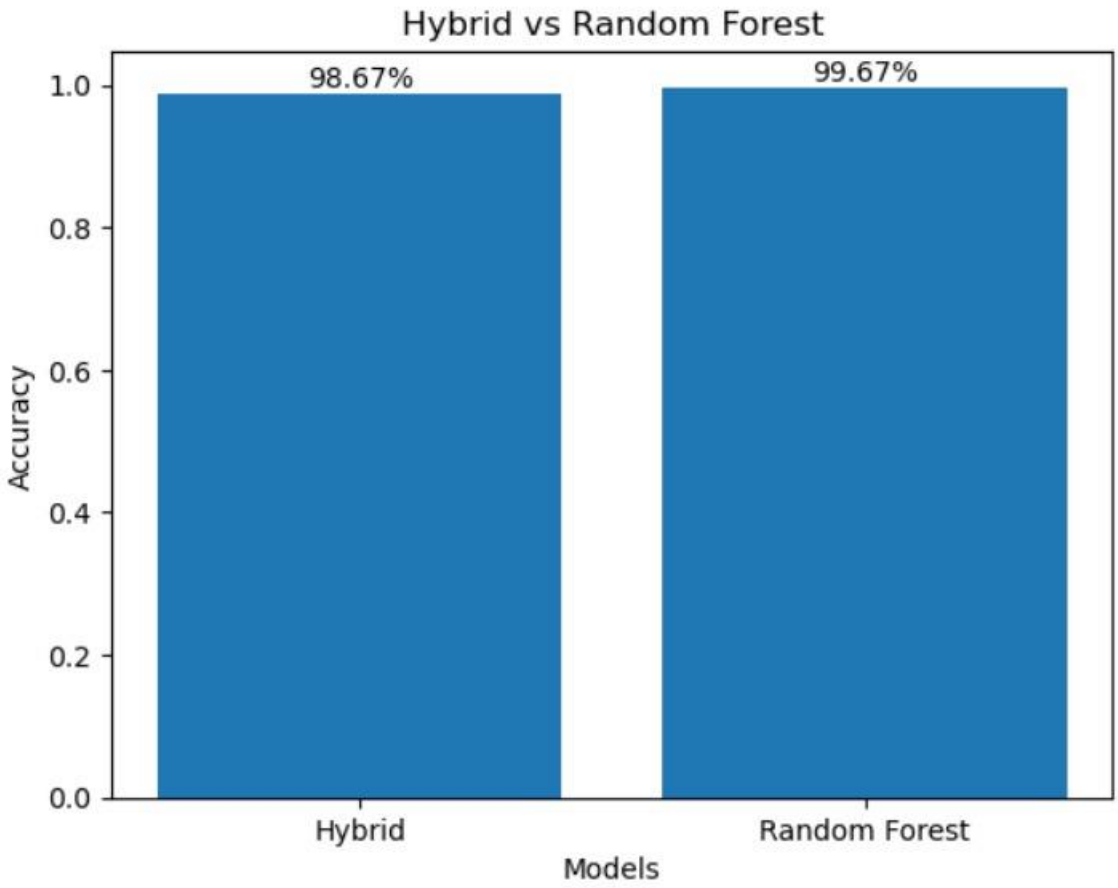


Fig 10.1 Comparative Analysis

11. Conclusion

The Intelligent Rain-Based Traffic Alert System demonstrates how IoT and machine learning can be combined to improve road safety during adverse weather conditions. The system uses simulated sensor data such as rain intensity, traffic presence, vehicle distance, temperature, humidity, visibility, and speed, which is stored in an Excel format and used to create a dataset for analysis and model training. Multiple machine learning models were trained to analyze the performance for traffic risk prediction. Support Vector Machine (SVM), Logistic Regression, and Decision tree classifier combined together as a Hybrid model has achieved accuracy of (98.67%) indicating improved performance in handling the dataset. Random Forest has achieved an accuracy of (99.67%) which shows that the data is overfitting. So, we conclude that the Hybrid model has provided the highest accuracy.